Refills and Renewals: Hassle Costs in the Contraceptive Market

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Abstract

In 2020, following the entry of a close competitor, Mirena, a leading hormonal intrauterine device (IUD), received its first extension of labeled duration after two decades on the market. Over the same period, 21 jurisdictions adopted laws allowing pharmacists to prescribe hormonal contraception. These developments reduced two forms of hassle costs: the burden of obtaining contraception and the frequency of returning to a provider to maintain protection. I estimate a nested logit model of contraceptive demand using Medicaid State Drug Utilization Data from 2015–2022. Counterfactual simulations show that extending Mirena's labeled duration by one year increases annual long-acting reversible contraceptive (LARC) insertions by approximately 14 percent and overall prescription contraceptive use by 2.8 percent. Nationwide pharmacist prescribing similarly increases use of eligible short-acting methods by approximately 7 percent and overall prescription contraceptive use by 4.1 percent. These responses demonstrate that reducing hassle costs induces substitution between methods while also expanding the prescription contraceptive market. In a further counterfactual, extending LARC durations to those supported by available clinical evidence generates approximately 226,500 additional annual insertions and averts an estimated 26,000 unintended pregnancies, yielding $31.8 million in projected annual net Medicaid savings after accounting for changes in contraceptive expenditures and the implied savings from avoided deliveries.

1 Introduction

The cost of accessing contraception extends well beyond out-of-pocket payment. Among reproductive-age women who report that they would use an over-the-counter contraceptive pill, 59 percent cite convenience as the primary reason, compared with only 6 percent who cite its potential to save money (Long et al., 2022). These responses reflect costs that are difficult to observe in claims but familiar to anyone seeking medical care, as obtaining and maintaining a prescription can require finding an available appointment, taking time away from work, arranging transportation or childcare, and returning to the health care system each time a prescription or device must be renewed. Nearly one in five reproductive-age women report that they are unlikely to be able to obtain their preferred contraceptive method on the same day they seek it, with distance to a provider, appointment availability, provider and pharmacy hours, transportation, and childcare among the barriers that can constrain timely access (Johnston et al., 2026). These costs create a nonfinancial margin of contraceptive access that persists even when the method itself is inexpensive or free to consumers, and one that varies considerably across methods according to how frequently a woman must interact with the health care system.

Between 2016 and 2022, 21 jurisdictions, including the District of Columbia, adopted policies allowing pharmacists to prescribe contraception, allowing women to obtain the pill, patch, and ring without first securing a prescription through a physician visit, while over the same period manufacturers extended the labeled years of protection of several long-acting reversible contraceptives, including Mirena's extension from five to eight years, increasing the time a woman could use an IUD or implant before returning to a provider for removal and replacement. These pharmacy access laws therefore reduce the hassle associated with obtaining contraception, removing the initial visit to a clinician, while longer labeled duration reduces the hassle associated with maintaining it, increasing the interval between insertion and replacement. Exploiting these two sources of variation, I ask how hassle costs shape contraceptive choice and how contraceptive use would change if long-acting methods provided the full duration of protection supported by clinical evidence.

I estimate contraceptive demand using Medicaid State Drug Utilization Data covering 48 states from 2015 to 2022. I construct state-year markets containing short-acting methods, implants, and IUDs and estimate a nested logit model in which women choose among these prescription methods and an outside option, which includes non-prescription contraception and non-use. Pharmacist prescribing authority is identified from the enactment of state pharmacy access laws, while labeled years of protection, which are chosen by manufacturers and may respond to product-specific demand, are instrumented using an indicator for whether a therapeutically equivalent hormonal-IUD rival was introduced or extended its approved duration in the preceding year and the number of close-dose rival hormonal IUDs. The resulting demand system recovers both women's preferences over these nonprice characteristics and the substitution patterns that determine where consumers move when the hassle associated with a method changes.

Women respond to reductions in both forms of hassle. Extending pharmacist prescribing nationwide increases use of eligible short-acting methods by approximately 7 percent (about 84,600 additional annual contraceptive choices). Of the approximately 84,600 additional eligible-method choices, approximately 70,800 come from the outside option, correspond-

ing to a 4.1 percent net increase in overall prescription contraceptive use. Longer labeled duration similarly increases demand for long-acting contraception. In a model simulation, extending the approved protection of Mirena, a leading hormonal IUD, by one year increases Mirena insertions by 56.9 percent, from approximately 140,000 to 220,000 annually, and increases total annual LARC insertions by approximately 14 percent, or 54,000 among Medicaid beneficiaries. Substitution is concentrated among clinically similar products, although the increase in total LARC insertions reflects an expansion of overall prescription contraceptive use by approximately 47,900 annual choices, or 2.8 percent, rather than simply substitution between devices. Together, the estimates show that reducing the frequency or difficulty of interacting with the health care system changes not only which contraceptive a woman chooses, but whether she uses prescription contraception at all.

Unlike pharmacist prescribing authority, however, the frequency with which a woman must return for replacement of a long-acting method is partly determined by the firm selling it. Between 2015 and 2022, four of the six LARCs marketed in the United States were at some point labeled to provide fewer years of protection than their existing clinical evidence could support, requiring replacement more frequently than the available evidence necessitated. Mirena entered the U.S. market in 2000 with a five-year label and remained there for nearly two decades despite evidence supporting seven years of protection as early as 1991 (Sivin et al., 1991). Following the 2015 entry of Liletta, a therapeutically similar hormonal IUD initially approved for three years of use, Bayer began the Mirena Extension Trial in December 2016 to evaluate its effectiveness beyond five years of use, while Liletta extended to four years in 2017, reached parity with Mirena at five years in 2018, and surpassed it with a six-year label in 2019. Mirena subsequently extended to six years in 2020, seven in 2021, and eight in 2022, when Liletta also reached eight years. Paragard and Nexplanon, each without a therapeutically equivalent rival during the sample, instead remained below the durations supported by their existing evidence. I consider a counterfactual in which Liletta never enters the market and assume Mirena remains at its original five-year approved duration. Relative to observed market outcomes, annual Medicaid LARC insertions fall by approximately 58,500, or 14.3 percent, while total prescription contraceptive use declines by approximately 52,000 annual choices, or 2.7 percent.

For women who wish to continue using these methods, shorter approved durations require more frequent replacement procedures, even when the underlying device could provide additional years of effective protection. The remaining gap between labeled duration and existing clinical evidence has consequences for contraceptive use, unintended pregnancy, and public spending. As a policy benchmark, I relabel each LARC to the maximum duration its accumulated evidence could support in each year of the sample, reducing the frequency of replacement without changing the underlying devices. These extensions would generate approximately 226,500 additional annual LARC insertions among Medicaid beneficiaries and increase overall prescription contraceptive use by approximately 201,000 annual choices, or 11.6 percent. Using a conservative assumption about pregnancy risk among women outside the prescription contraceptive market, this increase would avert approximately 26,000 unintended pregnancies and 12,500 births annually. Under a conservative fiscal calculation that charges Medicaid the full cost of each additional contraceptive device upon insertion but counts avoided delivery costs only in the year of insertion, these extensions generate approximately $31.8 million in projected annual net savings. This estimate excludes the

benefits of pregnancies prevented over the remaining years of each device's labeled lifespan, as well as potential savings from WIC and other longer-run public expenditures associated with unintended births.

A separate counterfactual examines nationwide pharmacist prescribing. Allowing women in every state to obtain eligible short-acting contraception directly from pharmacists increases use of these methods by approximately 84,600 annual contraceptive choices, while increasing overall prescription contraceptive use by approximately 70,800 annual choices, or 4.1 percent. The increase in overall use indicates that reducing the hassle of obtaining contraception can expand access beyond women who would otherwise have chosen a different prescription method. Accounting for the reduction in unintended pregnancies and associated delivery costs, the additional contraceptive spending is more than offset by savings to Medicaid, yielding approximately $2.46 million in annual net savings.

The consequences of contraceptive access extend beyond fertility to women's educational attainment and labor-market outcomes. By allowing women greater control over the timing of childbearing, contraception has altered decisions about education, marriage, and labor-market investment (Bailey, 2006; Bailey et al., 2012; Goldin & Katz, 2002). The consequences of reproductive control are also evident when pregnancy occurs earlier than intended. Miller (2011) estimates substantial earnings gains from delaying motherhood, using several sources of unexpected variation in fertility timing, including pregnancies occurring despite contraceptive use. More recently, Gallen et al. (2026) examine unplanned pregnancies among women using long-acting reversible contraception to delay childbearing and find that these pregnancies interrupt career progression, with persistent earnings consequences. Together, these studies demonstrate that the ability to achieve intended fertility timing has substantial value, especially for women who are already taking steps to avoid pregnancy. The consequences of unintended pregnancy therefore provide an important motivation for understanding not only whether women use contraception, but also which methods they choose and the efficacy those methods provide.

The extent to which women can exercise control over their fertility depends partly on the costs of obtaining contraception. A substantial literature examines how reducing financial barriers affects contraceptive use and fertility. Expansions of Medicaid family-planning eligibility increased access to subsidized contraception and reduced births, with evidence that greater contraceptive use accounts for the decline (Kearney & Levine, 2009). More recent initiatives have sought to increase access to long-acting reversible contraception, particularly among low-income women and adolescents. Colorado's Family Planning Initiative reduced teenage births, with especially pronounced effects among adolescents living near participating clinics (Kelly et al., 2020; Lindo & Packham, 2017). Experimental evidence similarly demonstrates that offering free contraception substantially increases both overall contraceptive use and adoption of long-acting reversible methods among low income women (Bailey, Lang, Vortuba, & Bartík, 2023). The effects of these initiatives nevertheless vary across settings. Virginia's Contraceptive Access Initiative reduced births more modestly, partly reflecting higher existing LARC use and differences in the populations served (Kiser et al., 2026). Together, these studies demonstrate that reducing financial barriers can increase contraceptive use and reduce fertility, while the magnitude of these responses depends on the setting and the extent of existing access. Financial coverage, however, does not eliminate the time and effort required to obtain contraception or maintain protection once a method

has been chosen.

Beyond monetary prices, contraceptive methods differ in the time and effort required to initiate and maintain use. Pennington and Venator (2024) develop a dynamic model of contraceptive choice in which women face monetary costs, preferences over method attributes, and nonpecuniary switching costs. They find that switching costs are substantial, helping explain persistence in contraceptive choice even when alternative methods become more attractive. Their analysis also demonstrates that expectations about future reproductive policies influence present contraceptive decisions, as changes in anticipated abortion access and contraceptive coverage alter the value of switching to more effective methods. Consistent with the importance of the reproductive-policy environment, Gallen and Lu (2026) find that the Dobbs decision increased sterilization and long-acting reversible contraceptive use in states hostile to abortion. Similarly, Restrictions on abortion providers demonstrate how greater travel distances and reduced provider availability can limit access to reproductive healthcare and increase births (Jones & Pineda-Torres, 2024; Lindo et al., 2020). These findings illustrate how contraceptive decisions depend on nonprice considerations, including the costs of switching methods and the broader policy environment in which reproductive choices are made.

Maintaining a method also requires recurring interactions with healthcare providers or pharmacies. Short-acting methods require prescriptions and refills, while long-acting methods eventually require replacement procedures. The frequency of these interactions varies across contraceptives, creating differences in the cost of continued use even when the products themselves are covered by insurance. Evidence from pharmacist-prescribing laws further illustrates the importance of these costs. Grossman et al. (2025) find that allowing pharmacists to prescribe hormonal contraception reduces births, indicating that the organization of contraceptive provision can affect reproductive behavior independently of changes in product prices. Their findings establish a fertility response to reducing the requirements for obtaining contraception, while leaving open how the policy changes demand across contraceptive methods and overall prescription contraceptive use.

Across healthcare markets, financial coverage and the availability of treatment do not necessarily ensure utilization when patients face administrative requirements, time costs, and other obstacles to obtaining care. Shepard and Wagner (2025) show that an additional enrollment requirement substantially reduces participation in subsidized health insurance. In a different healthcare setting, Gallegos et al. (2023) find that introducing a digital appointment system increases the use of preventive screening services, with reductions in transaction costs contributing to the response. Together, these studies illustrate how the requirements associated with obtaining healthcare can influence utilization independently of monetary prices. For contraception, the costs of navigating the healthcare system may be incurred repeatedly, as women obtain prescriptions, refill medications, or undergo replacement procedures. Because these interactions recur throughout women's reproductive years, reducing their frequency or the burden associated with each encounter offers a particularly promising opportunity to improve contraceptive access.

For long-acting contraception, the burden of maintaining protection is partly determined by a device's approved duration. This connects the analysis to a literature in industrial organization examining the relationship between product durability, consumer demand, and manufacturers' incentives. Swan (1970) and Swan (1971) examine firms' incentives to provide

durable products, while Bulow (1986) demonstrates how the prospect of future replacement sales can influence durability choices. This literature highlights a tension between the value consumers place on longer-lasting products and firms' incentives to generate repeat purchases, with the relationship depending on market structure and the nature of competition. In the contraceptive market, approved duration determines how frequently continuing users must replace a device, creating a related connection between consumer demand and replacement sales. I estimate how women value additional years of approved protection, examine suggestive evidence that competition may influence manufacturers' pursuit of duration extensions, and evaluate how contraceptive use might have differed under alternative market conditions and approved durations.

This paper contributes to the economic understanding of contraceptive access by examining how hassle costs shape not only whether women use prescription contraception, but also the methods they choose and the effectiveness of their contraceptive protection. Reducing the cost of obtaining or maintaining a particular method may induce substitution from another prescription contraceptive, changing the composition and effectiveness of protection without increasing overall prescription use. Alternatively, it may attract women who would otherwise rely on nonprescription methods or no contraception. The demand model distinguishes these responses, recovering women's valuation of pharmacist access and approved contraceptive duration while accounting for the endogeneity of the latter. In doing so, the analysis extends the literature on contraceptive access beyond financial affordability and the initial availability of a method to the recurring requirements associated with its use. It also complements existing evidence on the fertility consequences of pharmacist prescribing by identifying how changes in access alter demand across contraceptive methods and the outside option that differ in efficacy.

The analysis further connects contraceptive demand to the economics of product durability by examining how competition may influence the approved duration of contraceptive devices and the consequences of those decisions for consumers. I estimate women's valuation of additional years of approved protection and examine the historical pattern of duration extensions among competing hormonal IUDs, which provides suggestive evidence that competition may influence manufacturers' incentives to pursue longer approved durations. I then use the estimated demand model to evaluate how contraceptive use might have differed in the absence of Liletta's entry, under the assumption that Mirena remained at its original five-year approved duration, and how use would change if existing devices were approved for the full duration supported by clinical evidence. Together, these analyses illustrate how the interaction between market structure and approved product duration can affect contraceptive choice and overall prescription contraceptive use.

The consequences of these demand responses extend beyond the contraceptive market. The labor-market consequences of unintended pregnancy documented by Miller (2011) and Gallen et al. (2026) arise partly because women may experience pregnancies at times that disrupt their intended educational, professional, or family decisions. Changes in contraceptive access can affect the likelihood of such pregnancies through both the decision to use prescription contraception and the effectiveness of the methods women select. I use the estimated demand responses, together with assumptions about contraceptive failure and pregnancy risk, to calculate the implied changes in unintended pregnancies, births, and Medicaid expenditures. Although these calculations do not measure the longer-run consequences

for women's economic outcomes, they connect the costs of obtaining and maintaining contraception to the reproductive events that can shape those outcomes. More broadly, the analysis speaks to general healthcare access by demonstrating how the burden of repeated interactions with providers can influence the use of treatments whose benefits must be maintained over time. For contraception, the distinction between obtaining treatment and maintaining its benefits is particularly stark: a method may be financially accessible, yet the time and effort required to continue using it can determine whether a woman remains protected against an unintended pregnancy. Reducing these recurring burdens therefore offers a means of improving healthcare access that extends beyond lowering prices, with implications for both the utilization of medical care and the outcomes that care is intended to prevent.

2 Background

2.1 Contraception in the United States

The modern contraceptive market offers substantial product differentiation across multiple dimensions. Prescription contraceptives, summarized in Tables 1 and 2, include short-acting methods such as pills, patches, rings, and injections and long-acting reversible contraceptives, or LARCs, which include implants and intrauterine devices. Short-acting methods vary in administration frequency from daily pills and weekly patches to monthly rings and injections administered every three months. LARCs instead provide several years of protection following a single insertion and are reversible, meaning that the device can be removed before the end of its approved period of use. Hormonal and copper IUDs are inserted into the uterus, while the contraceptive implant is inserted under the skin of the upper arm. Within the LARC category, the approved period of protection varies substantially across products, ranging during the sample from three years to ten years of coverage.

These differences in administration frequency also determine how often obtaining or maintaining a method requires interaction with the health care system. Pills, patches, and rings are self-administered but must be obtained repeatedly and traditionally required a prescription from a clinician, while implants and IUDs require a clinical procedure for insertion and another for removal or replacement. Once an IUD or implant has been inserted, an additional approved year allows the same device to remain in place for another year before a woman who wants to continue using it must return for replacement.

Despite the variety of contraceptive methods, most categories are highly concentrated. Combination pills are the exception, with roughly 180 products from 26 manufacturers, while the contraceptive implant and copper IUD each have only one product on the market, Nexplanon and Paragard respectively, and three of the four hormonal IUDs belong to a single manufacturer, with Skyla, Kyleena, and Mirena all produced by Bayer. Liletta, the remaining hormonal IUD, is co-developed and co-commercialized by Allergan and Medicines360, a nonprofit pharmaceutical organization.[1] In particular, Mirena and Liletta are therapeutic substitutes, containing the same 52-mg dose of levonorgestrel and offering the same effi-

[1] Allergan and Medicines360 co-developed and co-commercialize Liletta, with responsibilities divided by care setting. Medicines360 serves the public and safety-net channel, while Allergan serves the commercial channel.

cacy rates, while Kyleena and Skyla contain lower hormonal doses, Paragard contains no hormones, and Nexplanon differs both in hormone and route of administration.

Differences in financial costs between these products are small to Medicaid beneficiaries. Under the Affordable Care Act's contraceptive coverage requirements, participating Medicaid expansion programs cover prescription contraception with little or no patient cost sharing. In the data, approximately 84 percent of recorded patient-side payments are below $10 and 77 percent are below $5, with roughly 75 percent of LARC payments and 93 percent of pill payments below $10. These recorded payments can include contributions from secondary payers such as Title X and therefore provide an upper bound on what the woman pays directly.[2]

2.2 Pharmacy Access Laws

Although pills, patches, and rings are dispensed at pharmacies and administered by the woman herself, obtaining these methods traditionally required a prescription from a health care provider. Beginning in 2016, states increasingly allowed pharmacists to prescribe these methods directly, eliminating the need to obtain the prescription through a separate clinician visit. Twenty-one states adopted pharmacist prescribing authority during the 2016–2022 sample period, with five states adopting in 2016, three in 2017, one in 2018, four in 2019, one in 2020, four in 2021, and three in 2022.[3]

Pharmacist prescribing changes the process required to obtain self-administered contraception rather than the contraceptive product itself. Under these laws, women can obtain prescriptions for eligible methods directly from a pharmacist rather than first obtaining a prescription from a clinician. The laws apply to the pill, patch, and ring, although their specific requirements vary across states.

The clinician visit required to obtain a prescription can impose both scheduling and travel costs. Among publicly supported family-planning clinics, the average wait for an initial contraceptive appointment increased from three days in 2015 to six days in 2022–2023, while the share offering same-day appointments fell from 52 to 25 percent (VandeVusse et al., 2024). Community pharmacies are also considerably more geographically accessible, with 89 percent of the U.S. population living within five miles of a pharmacy (Berenbrok et al., 2022). Pharmacist prescribing therefore reduces the clinical interactions required to initiate or continue an eligible method and shifts contraceptive prescribing toward a more geographically available setting.

2.3 Years of Protection and Clinical Evidence

For a LARC, FDA approval specifies the number of years for which the device may be marketed as providing contraceptive protection. Increasing this period requires the manufacturer

[2]Title X is a federally funded program dedicated to the provision of low-cost family planning and reproductive health care.

[3]California, Colorado, Oregon, Virginia, and Vermont adopted pharmacist prescribing authority in 2016; Hawaii, New Hampshire, and New Mexico in 2017; the District of Columbia in 2018; Idaho, Maryland, Utah, and West Virginia in 2019; Minnesota in 2020; Arizona, Arkansas, Delaware, and Nevada in 2021; and Illinois, North Carolina, and South Carolina in 2022.

to submit evidence supporting effective use over the additional years, but does not require a change to the device itself. Liletta entered the market in 2015 with three years of approved protection and increased to four years in 2017, five in 2018, six in 2019, and eight in 2022. Mirena entered the U.S. market in 2000 with five years of protection and remained at five through 2019 before increasing to six years in 2020, seven in 2021, and eight in 2022. Kyleena remained at five years, Skyla at three, Paragard at ten, and Nexplanon at three throughout the periods in which they appear in the sample.

The clinical evidence available to a manufacturer can support additional years of use before the firm seeks FDA approval to market the product for those years. I define the longest period supported by the evidence available in each year as the product's feasible years of protection, with the approved and feasible periods shown in Table 3. For Liletta, the relevant evidence comes from the ACCESS IUS trial, which enrolled approximately 1,700 women between 2009 and 2013 to evaluate the 52-mg levonorgestrel device for as long as ten years. I treat an additional year as feasible once approximately 350 trial participants have completed that year of use, corresponding to the smallest number of completers underlying Liletta's actual approvals. This places Liletta's feasible protection at four years in 2015, five in 2016, six in 2017, seven in 2018, and eight beginning in 2019, leaving the firm with evidence supporting additional years before those years appeared on the product's FDA-approved label.

Mirena entered the U.S. market in 2000 with five years of approved protection and remained there for nearly two decades. Evidence of contraceptive effectiveness beyond five years, however, substantially predates Mirena's eventual extensions. A randomized study published in 1991 found that a levonorgestrel-releasing IUD provided effective contraception through seven years of use (Sivin et al., 1991). Subsequent research examining extended use of Mirena documented continued levonorgestrel release and endometrial effects beyond the approved five-year period, although it did not establish sufficient evidence to recommend routine extended use (Hidalgo et al., 2009). Following the entry of its therapeutic substitute, Liletta, in February 2015, Bayer began the Mirena Extension Trial on December 22, 2016, enrolling existing Mirena users who were already 4.5 to 5 years into use. Because participants entered the trial near the end of the originally approved five-year period, efficacy data for each successive year became available roughly one year apart, with data supporting six years of use accumulating through 2017, seven years through 2018, and eight years through 2019. I therefore treat six years as feasible for Mirena beginning in 2018, seven in 2019, and eight in 2020. Mirena's approved protection remained at five years through 2019 before increasing to six in 2020, seven in 2021, and eight in 2022.

Evidence supporting longer use was also available for Paragard and Nexplanon, neither of which increased its approved years of protection during the sample. Paragard has carried a ten-year approval since 1994, although evidence of effectiveness through twelve years has existed since a large multicenter trial reported continued efficacy in years eleven and twelve in 1997, with later studies providing additional evidence of extended use. Nexplanon remained approved for three years throughout the sample, although studies conducted before the sample reported no pregnancies among women continuing implant use through years four and five.[4] Kyleena and Skyla, in contrast, remain at the protection periods supported by

[4]The extended-use evidence for Nexplanon comes from observational cohort data with substantially fewer

their evidence throughout the sample.

As shown in Table 3, four of the six LARCs had evidence supporting more years of use than they were approved to market at some point between 2015 and 2022. Table 3 reports the approved and feasible years of protection over the sample, illustrating both the repeated increases for Mirena and Liletta and the persistent gap for Paragard and Nexplanon.

This distinction between approved and feasible duration matters as the approved period determines when women are instructed to return for replacement, even when clinical evidence suggests that the same device could continue providing effective protection. For women who wish to continue using a LARC, a shorter approved duration requires more frequent replacement procedures and the associated time and effort. If women value longer protection, extending approved duration can therefore increase the attractiveness of a method without changing the underlying device. The gap between approved and feasible duration consequently represents an opportunity to reduce the recurring costs of contraception through longer periods of approved use.

3 Conceptual Framework

Contraceptive methods differ not only in price, efficacy, and side effects, but also in the time and effort required to maintain protection. These recurring costs arise through physician visits, prescription renewals, pharmacy trips, and replacement procedures. Although they are not necessarily reflected in the monetary price of contraception, they enter the consumer's choice of method and may affect whether she uses prescription contraception at all. Two features of the contraceptive market make these costs particularly relevant. Short-acting methods require repeated interactions to maintain use, while long-acting methods require fewer interactions but eventually must be replaced.

Consider a woman choosing among contraceptive methods indexed by j, including an outside option representing nonprescription contraception or no prescription method. Her utility from method j can be written as

$$U_{ij} = v_{ij} - p_{ij} - h_{ij},$$

where v_{ij} captures the value of contraceptive protection and other method characteristics, p_{ij} is the monetary cost of the method, and h_{ij} represents the nonmonetary costs of obtaining and maintaining it. These costs include the time and effort associated with obtaining prescriptions, refilling medication, and undergoing replacement procedures. The relevant cost is not simply the inconvenience of an individual interaction, but also how frequently that interaction must occur over the period of contraceptive use.

For short-acting methods, the recurring cost depends on the frequency of prescription renewals and refills, as well as the requirements governing those interactions. Pharmacist prescribing reduces the need to obtain prescriptions through a physician, lowering the cost of maintaining eligible methods without changing their contraceptive efficacy or physical characteristics. For long-acting methods, recurring costs depend in part on approved duration. An additional year of protection extends the interval before replacement is required,

completers than the threshold applied to Liletta, rather than from a manufacturer extension trial.

reducing the frequency of procedures and the associated time and effort. These changes operate through different institutional mechanisms but affect the same underlying component of contraceptive utility.

A reduction in recurring costs increases the utility of the affected method relative to its alternatives. The resulting increase in demand can arise from substitution among prescription contraceptives or from women choosing a prescription method instead of the outside option. The distinction depends on preferences and substitution patterns across methods. If women regard the affected method and other prescription contraceptives as close substitutes, much of the demand response may reflect reallocation within the market. If the reduction in recurring costs makes prescription contraception preferable to the outside option for some women, total prescription contraceptive use increases.

The framework highlights two ways in which the costs of maintaining contraception can change. Pharmacist prescribing reduces the burden of obtaining short-acting methods, while longer approved durations reduce the frequency of replacement procedures for long-acting methods. Although these changes operate through different mechanisms, both reduce the recurring costs associated with contraceptive use. By reducing the time and effort required to maintain protection, these changes increase the attractiveness of prescription contraception relative to alternatives that require fewer ongoing interactions.

4 Data

I combine Medicaid prescription drug utilization with enrollment and demographic data to construct state-year contraceptive markets from 2015 through 2022. The primary data source is the Medicaid State Drug Utilization Data (SDUD), supplemented with Medicaid enrollment from Kaiser Family Foundation (KFF) compilations of the Transformed Medicaid Statistical Information System (T-MSIS) and state enrollment reports (Foundation, 2023), which together provide enrollment counts across the full sample period. I also use demographic information from the Current Population Survey (CPS) and contraceptive-use histories from the National Survey of Family Growth (NSFG).

4.1 Medicaid Data and Enrollment

The SDUD contain information on covered outpatient drugs paid for by state Medicaid programs. The data are reported quarterly and separately for fee-for-service and Medicaid managed care, which I aggregate to the state-year level to match the frequency and level at which Medicaid enrollment is observed.[5] I exclude Washington, South Dakota, and Tennessee because of systematic reporting inconsistencies, leaving 48 state and District of Columbia markets in the analysis.[6] For each contraceptive identified by its National Drug Code (NDC),

[5]Under fee-for-service Medicaid, the program reimburses providers directly, while under managed care Medicaid pays a capitated amount to a managed care organization. Because the enrollment data do not separately identify enrollment in fee-for-service and managed care, I aggregate utilization across the two systems within each state-year.

[6]Preliminary examination of the SDUD revealed implausible utilization figures, including more reported IUD prescriptions in South Dakota than residents of the state. Schwartz et al. (2016) similarly document anomalously high Medicaid prescription spending in Washington, South Dakota, and Tennessee in an earlier

the SDUD report the number of prescriptions and units reimbursed, the amount reimbursed by Medicaid, payments from non-Medicaid sources, and total reimbursement. These measures apply to the contraceptive product itself and do not include payments for associated clinical services such as insertion, removal, or consultation. Appendix A describes the complete construction and cleaning of the analysis data.

I combine KFF compilations of T-MSIS with state Medicaid enrollment reports to obtain total Medicaid enrollment for each state-year, using the two sources to cover the entire sample period. Neither source consistently reports the number of female enrollees of reproductive age. I use the CPS Annual Social and Economic Supplement to estimate the share of Medicaid enrollees who are women ages 18 to 45 and apply that share to total enrollment.[7]

4.2 Product Construction

The unit of observation is a state-year-product. I begin with individual NDCs and aggregate formulations that represent similar contraceptive products while preserving distinctions in characteristics likely to affect contraceptive choice. Oral contraceptives are grouped into five categories according to estrogen content, consisting of progestin-only pills and four estrogen-dose groups ranging from ultra-low to high dose. Rings, patches, and injections are each aggregated into a single product category. I retain implants and IUDs at the brand level, preserving Mirena, Liletta, Kyleena, Skyla, Paragard, and Nexplanon as separate products as they differ in years of protection, hormonal composition, route of administration, and other clinical characteristics.

Years of protection are assigned according to the FDA-approved period in force for each product and year. Short-acting methods receive fractional-year values corresponding to the period of protection provided by each administration, with 1/365 for daily pills, 1/52 for weekly patches, 1/12 for monthly rings, and 1/4 for quarterly injections. IUDs and the implant receive their product-specific approved years of protection, including changes in Mirena's and Liletta's approved protection over the sample. The remaining characteristics entering the demand model are amenorrhea and pharmacist access. Amenorrhea indicates whether a method can result in the absence of menstrual bleeding, with method-specific rates drawn from the clinical evidence summarized in Table 2. Pharmacist access varies by product, state, and year according to whether state law permits pharmacist prescribing of the method.

4.3 Markets and Choice Shares

A market is defined as a state-year, with the initial market size equal to the estimated number of female Medicaid enrollees ages 18 to 45. Women already using a LARC remain protected for multiple years and are therefore not in the market for a new contraceptive in every year. Since the aggregate data do not identify individual women or when current enrollees received a LARC, I approximate this stock following Armitage (2022). I use historical LARC insertion shares to estimate the share of women in the current market who received a LARC

period.

[7] As a robustness check, I compare these estimates with states that directly report Medicaid enrollment by age and sex.

in an earlier year and remain protected by that device. This construction assumes that the contraceptive histories of women in the current Medicaid population resemble those of women observed in the same state in preceding years. I subtract this estimated stock of continuing LARC users from the number of female Medicaid enrollees to obtain the population making a contraceptive choice in each state-year. Continuing LARC users account for approximately 13 million women across these markets, reducing the aggregate market size by 11.83 percent. Appendix A describes the construction in detail.

Product shares approximate the fraction of women choosing each contraceptive in a state-year, so the prescription and unit counts are converted to reflect the number of women using a year's supply of each product in a market. For pills, I divide the total number of pills reimbursed by 365 to approximate the number of women supplied with one year of pills. For patches and monthly rings, I divide the number of prescriptions by 12, reflecting monthly dispensing. For injectable contraception, I divide the units reimbursed by the standard 150 mg dose to recover the number of injections. Each IUD or implant prescription corresponds to a new device and is therefore counted as one insertion. I divide these quantities by the market size in the state-year to obtain each product's market share.

The outside option is the residual share of women not observed receiving one of the prescription contraceptives in the SDUD. It therefore includes non-use as well as contraceptive methods not observed in the prescription drug data, including condoms and other over-the-counter methods, emergency contraception, fertility-awareness methods, and permanent contraception.

The NSFG provides complementary information on transitions across contraceptive methods that is not available in the aggregate Medicaid data. I use contraceptive histories from the 2011–2019 NSFG to examine which methods women choose after discontinuing a prior method. I also use the NSFG to measure contraceptive efficacy for the outside option used in counterfactual exercises.

4.4 Reimbursement and Patient-Side Payments

The SDUD report Medicaid reimbursement, reimbursement from non-Medicaid sources, and total reimbursement for each NDC. Total reimbursement measures revenue generated by the contraceptive product itself, excluding associated clinical services such as insertion, removal, or consultation. Patient-side payments are constructed from the non-Medicaid reimbursement field. This field can include payments from secondary sources in addition to patient cost sharing and therefore provides an upper bound on the amount paid directly by the woman.[8]

A small number of observations contain implausible payments. I identify these observations using method-specific thresholds based on the maximum annual cost of each contraceptive as reported by Planned Parenthood (Planned Parenthood, 2019). I replace observations with payments that exceed the corresponding maximum annual cost using the payment from the closest year that falls below this ceiling for the same state and product, followed by the product-level median when no such observation is available. Prices are imputed for 4.31 per-

[8]Non-Medicaid payments can include payments from other sources, including Title X family-planning funds.

cent of state-year-product observations, including less than 1 percent of LARC prescription volume.

Medicaid generally covers family-planning services and supplies without cost sharing. Additionally, under the ACA, Medicaid expansion coverage must also include at least one contraceptive product in each FDA-approved method category with zero cost sharing to the patient. Consistent with these requirements, patient-side payments are small. Approximately 84 percent of observations are below $10 and 77 percent are below $5, with 75 percent of LARC payments and 93 percent of pill payments below $10.

4.5 Descriptive Evidence

Table 4 describes contraceptive use over the sample period. The outside option accounts for approximately 82–85 percent of the market, while short-acting methods account for approximately 12–14 percent and LARC insertions approximately 3–4 percent. Among observed contraceptive choices, pills account for the largest share of use. Table 5 reports product-level market shares, within-nest shares, and reimbursement in the estimation sample.

The predominance of self-administered short-acting contraception makes the expansion of pharmacy access relevant for a large share of observed contraceptive use. Pharmacy access expands from no jurisdictions at the beginning of the sample to 21 by 2022, with states adopting throughout 2016–2022 (Figure 1). These laws allow pharmacists to prescribe the pill, patch, and ring, eliminating the separate clinician visit otherwise required to obtain a prescription for these methods. These methods account for approximately one-eighth of the contraceptive market throughout the sample, with short-acting self-administered products being the most commonly used group in all years, driven largely by the pill. The mean share of these self-administered products is about 12.60 percent in state-years with access as compared to 12.47 in those without (Table 6). Among jurisdictions that adopt during the sample, eligible methods account for 11.68 percent of the market before adoption and 11.88 percent after adoption (Table 6).

Approved years of protection vary both across LARC products and within products over time. Table 3 reports approved years of protection for the four hormonal IUDs, while Figure 2 shows their shares of hormonal-IUD insertions. Mirena enters the sample with five approved years of protection and remains at five years through 2019 before extending to six, seven, and eight years between 2020 and 2022. Liletta enters with shorter approved protection and extends repeatedly during the sample, while Kyleena and Skyla remain at five and three years, respectively. Within the same period, the composition of hormonal-IUD use changes substantially (Figure 2). Liletta gains share as its approved protection increases, while Mirena remains the dominant hormonal IUD but loses share over the sample.

I examine this relationship using an indicator for whether a therapeutic equivalent introduced or extended its approved protection in the preceding year. The introduction of a competing product or an extension of its approved protection changes the competitive environment, potentially increasing the incentive for a manufacturer to extend its own product's duration. In the first stage of the demand estimation, I regress approved years of protection on this indicator, controlling for method, state, and year fixed effects, amenorrhea, pharmacist prescribing, and the availability of close-dose IUD competitors. A rival's introduction or extension in the preceding year is associated with 0.48 additional years of approved protec-

tion (Table 7). This positive relationship is consistent with manufacturers' duration choices responding to changes in the protection offered by competing products.

Neither this relationship nor the association between approved protection and utilization identifies consumers' response to longer protection. Manufacturers choose whether and when to pursue extensions, and those choices may themselves respond to product demand. I address this endogeneity in the demand model below.

5 Demand Model

The descriptive evidence shows how utilization covaries with pharmacy access and approved years of protection, but these relationships do not identify women's valuation of either attribute. I estimate a nested-logit model of contraceptive demand in which women choose among fourteen contraceptive products and an outside option.

The products are grouped into three nests based on their clinical characteristics. The four hormonal IUDs form one nest, the copper IUD and implant form a second LARC nest, and the short-acting methods form a third. The short-acting nest contains the injection, ring, patch, and five pill categories defined by estrogen dose: progestin-only, ultra-low-dose, low-dose, standard-dose, and high-dose pills. The nesting structure implies stronger substitution among products within the same clinical group than among products in different groups.

The outside option includes any over-the-counter method, such as condoms or spermicide, along with emergency contraception, abortion, the rhythm method, sterilization, and abstention or non-use. I model contraceptive choice within each state-year as static, abstracting from forward-looking decisions about fertility and future contraceptive use. Women estimated to remain protected by a LARC inserted in an earlier year are excluded from the current market size for the remaining years of approved protection, as described in Section 4.

Woman i's utility from product j in nest g in market st is

$$U_{ijst} = \delta_{jst} + \zeta_{ig} + (1-\rho)\epsilon_{ijst},$$

where δ_{jst} is the mean utility of product j in market st, ζ_{ig} is a woman-specific preference for nest g, and ϵ_{ijst} is an idiosyncratic product-specific taste shock. I assume ϵ_{ijst} is independently distributed type-I extreme value across products and ζ_{ig} is distributed such that the combined error $\zeta_{ig} + (1-\rho)\epsilon_{ijst}$ is also type-I extreme value. The parameter ρ, with $0 \leq \rho < 1$, is the nesting parameter, with larger values corresponding to stronger correlation in preferences among products in the same nest. Utility from the outside option is normalized to zero.

Each product's mean utility in market st, δ_{jst}, is decomposed as

$$\delta_{jst} = \beta D_{jt} + \phi R_{jst} + \gamma A_j + \mu_{m(j)} + \mu_s + \mu_t + \xi_{jst},$$

where D_{jt} is approved years of protection, R_{jst} indicates that an eligible product can be prescribed by a pharmacist in the state-year, and A_j is an indicator for products with high rates of amenorrhea. Method fixed effects $\mu_{m(j)}$ distinguish pills, patches, rings, injections, implants, hormonal IUDs, and copper IUDs. State and year fixed effects absorb persistent differences across states and changes common across markets in a given year. The term ξ_{jst}

contains product-market characteristics observed by women but unobserved by the econometrician that affect mean utility. The coefficient β measures the marginal utility of an additional approved year of protection, and ϕ measures the change in utility for eligible products when pharmacist prescribing becomes available.

Given that out-of-pocket prices faced by Medicaid enrollees are zero or near zero, out-of-pocket payment does not enter utility[9]. As shown above, roughly 84 percent of recorded contraceptive payments are below $10, including 75 percent for LARCs and 93 percent for pills.

Following Berry (1994), the nested-logit model can be inverted to obtain

$$\ln(s_{jst}) - \ln(s_{0st}) = \beta D_{jt} + \phi R_{jst} + \gamma A_j + \rho \ln(s_{j|g,st}) + \mu_{m(j)} + \mu_s + \mu_t + \xi_{jst},$$

where s_{jst} is product j's market share, s_{0st} is the outside-option share, and $s_{j|g,s,t}$ is product j's share among products in its nest. I impose a common nesting parameter ρ across the three nests as a simplifying restriction on substitution patterns. I estimate this equation by two-stage least squares, instrumenting for approved years of protection and within-nest share. Standard errors are clustered by market.

Pharmacy access is identified from the adoption of state laws allowing pharmacists to prescribe self-administered contraception. I define

$$R_{jst} = Eligible_j \times L_{st},$$

where $Eligible_j$ equals one for pills, patches, and rings, and L_{st} indicates that pharmacist prescribing is available in state s and year t. Considering a state that adopts pharmacy access in 2019, ϕ is identified from the change in demand for eligible methods following adoption, relative to demand for those methods in state-years in which pharmacy access is not available. Method fixed effects absorb persistent differences across contraceptive methods, state fixed effects absorb persistent differences across states, and year fixed effects absorb changes common across states and methods in a given year. Identification requires that the timing of adoption not coincide with other changes that differentially affect demand for the pill, patch, and ring.

Approved years of protection are potentially endogenous if manufacturers' decisions to pursue extensions respond to demand conditions. I instrument for years of protection using an indicator for whether a therapeutic equivalent introduced or extended its approved protection in the preceding year. The indicator equals one when either product's competitor entered the market or received approval for longer protection in the preceding year, and zero for all other products. A rival's introduction or extension can increase the incentive to pursue an extension by changing the relative protection offered by competing products. For example, Liletta's introduction and subsequent extensions reduced Mirena's advantage in approved protection over a close competitor, potentially increasing Bayer's incentive to pursue an extension in the following period. The descriptive evidence in Section 4.5 is consistent with this relationship, showing that a therapeutic equivalent's introduction or extension predicts longer approved protection in the following year.

[9]The estimates are robust to including out-of-pocket payment as an additional control (Appendix Table 14).

The identifying assumption is that a therapeutic equivalent's introduction or extension in the preceding year is uncorrelated with unobserved determinants of current demand for the product whose duration is being instrumented. A rival's introduction or extension changes the competitive environment facing the manufacturer and may influence its decision to pursue longer approved protection. The lag establishes this competitive environment before current demand is realized, and I assume the instrument affects current demand only through its relationship with the product's own approved duration. A potential violation would arise if Liletta's introduction or extension increases demand for Liletta at Mirena's expense for reasons unrelated to their observed clinical characteristics. For example, women might develop a preference for Liletta as a brand, or providers might begin recommending or stocking Liletta instead of Mirena independently of differences in approved protection, hormone dose, or other product attributes. Brand-specific preferences may be less important in this setting as women choose among hormonal IUDs that are differentiated by clinically relevant characteristics, including hormone dose, device size, and approved years of protection, rather than by brand alone. Similarly, providers' recommendations are likely to reflect these clinical characteristics, so changes in recommendations following a competitor's extension may largely reflect differences in attributes already incorporated into the demand model. Finally, inertia from previous choices is also less likely to generate persistent annual demand for a particular brand because women typically make a new IUD choice only at removal or replacement several years later.

I instrument for within-nest share using the number of other hormonal IUDs with a similar hormone dose that are available in each state-year. The availability of a close substitute affects how demand is divided among products within the nest, providing variation in within-nest shares. This approach follows the use of rival product characteristics as instruments in differentiated-products demand models (Berry, 1994). I divide hormonal IUDs into high-dose and lower-dose products, with Mirena and Liletta in the high-dose group and Kyleena and Skyla in the lower-dose group, and count the number of other products in the same group. The identifying variation comes from the diffusion of Liletta and Kyleena across state Medicaid markets following their introduction in 2015 and 2016, respectively. I treat each product as available beginning in the first year in which it appears in a state's Medicaid claims and thereafter. For example, Liletta's appearance in a state increases the number of close competitors facing Mirena and should reduce Mirena's share within the hormonal-IUD nest. The exclusion restriction assumes that the state-level availability of a close competitor is uncorrelated with unobserved demand for the focal product, conditional on the included controls (Berry, 1994).

Table 7 reports the first-stage estimates for approved protection and within-nest share. The excluded instruments strongly predict both endogenous variables conditional on the included controls and one another. A therapeutic equivalent's introduction or extension in the preceding year is associated with 0.48 additional years of approved protection, consistent with manufacturers responding to changes in the competitive environment by pursuing longer approved protection for their own products. The Sanderson–Windmeijer conditional first-stage F-statistics are 405 for approved protection and 112 for within-nest share, indicating strong instrument relevance for both endogenous variables after accounting for their joint determination (Sanderson & Windmeijer, 2016). In the within-nest-share first stage, the availability of an additional close IUD competitor is associated with a lower share for the

product within its nest. This negative relationship is consistent with the substitution mechanism underlying the instrument: the entry of a close alternative divides demand among more products within the hormonal-IUD nest, reducing the share captured by an existing product.

6 Demand Results

The results show that women place substantial value on reducing the hassle costs of both maintaining and obtaining contraception. Women prefer methods that provide longer protection before another clinical visit is required, while removing the clinician-prescription requirement increases demand for short-acting methods.

Table 8 reports a coefficient on approved years of protection of 0.314 (SE 0.149), statistically significant at the 5 percent level, while the nesting parameter of 0.743 (SE 0.099) is statistically significant at the 1 percent level and indicates substantial substitution among clinically similar products. To interpret the magnitude of the duration estimate, I calculate the average marginal effect of an additional year of approved protection from each LARC's observed duration. An additional year increases a hormonal IUD's share within its nest by 12 to 17 percentage points, corresponding to a 0.2 to 0.8 percentage-point increase in its share of the contraceptive market as a whole, depending on the IUD's baseline duration and market share. The effects are similar in the other LARC nest, where an additional year increases the whole-market share of the copper IUD by approximately 0.6 percentage points and the implant by 0.8 percentage points. These estimates isolate the value of extending protection without changing the underlying device, showing that the frequency with which a method requires another clinical interaction is itself an important determinant of contraceptive choice.

The pharmacy access estimates provide complementary evidence that the process required to obtain contraception affects demand even when the underlying product remains unchanged. Table 8 reports a pharmacist-prescribing coefficient of 0.084 (SE 0.049), statistically significant at the 10 percent level, with universal pharmacist prescribing increasing the combined market share of pills, patches, and rings by 0.79 percentage points, from 12.50 to 13.29 percent, an increase of approximately 7 percent in eligible-method use. Most of this additional demand comes from the outside option, which falls by 0.67 percentage points and accounts for 83.8 percent of the increase in eligible methods. The injection accounts for another 0.09 percentage points, while the hormonal IUDs, implant, and copper IUD together account for only 0.04 percentage points. Reducing the time and effort required to obtain a prescription therefore affects whether women use prescription contraception at all, rather than simply changing which prescription method they choose.

Together, these results show that contraceptive choice depends not only on the clinical characteristics of a method, but also on the process required to obtain and continue using it. This distinction matters for understanding demand in health-care markets more broadly, where the time, effort, and frequency of interactions required to use a product can be economically important attributes even when they are not reflected in its clinical characteristics. For contraceptive policy, the estimates imply that making a method available is distinct from making it easy to obtain and maintain.

6.1 Demand Counterfactuals

The demand estimates show that reducing the hassle associated with obtaining and maintaining contraception increases demand. I use the model to quantify how contraceptive use would change if the remaining frictions along each margin were reduced. The first counterfactual extends pharmacist prescribing to every state, while the second sets each LARC's approved years of protection equal to the maximum that could have been supported by the clinical evidence available in each year. In both exercises, I hold other product characteristics and unobserved demand shocks fixed and allow women to reoptimize among contraceptive methods and the outside option.

Extending pharmacy access to every state increases the combined share of pills, patches, and rings from 12.50 to 13.29 percent, a 0.79 percentage-point increase. Most of this increase comes from the outside option rather than substitution from other prescription methods, increasing total prescription contraceptive use from 16.95 to 17.62 percent. Applied to the Medicaid population represented by the estimation sample, this increase corresponds to approximately 70,800 additional annual prescription contraceptive choices. Universal pharmacy access therefore increases prescription contraceptive use by approximately 4.1 percent relative to baseline. In addition to reducing the acquisition costs faced by women, pharmacist prescribing allows routine contraceptive prescribing to move outside the clinical setting, potentially freeing appointment capacity for care that requires a clinician.

The second counterfactual examines the maintenance costs associated with LARCs (Table 9). I set each product's approved duration equal to the feasible protection reported in Table 3, which reflects the clinical evidence available in each year rather than imposing the protection periods ultimately approved by the FDA throughout the sample. Moving each LARC to its feasible duration increases the combined LARC share from 3.90 to 6.12 percent, a 2.22 percentage-point increase, or approximately 58 percent relative to baseline. Total prescription contraceptive use increases from 16.95 to 18.83 percent, approximately 11.6 percent, as most of the additional LARC use comes from women who would otherwise choose the outside option. The counterfactual generates approximately 226,500 additional LARC insertions in an average year. The response is substantially larger than under universal pharmacy access and indicates that the frequency with which continued contraceptive protection requires another clinical interaction can materially affect both the choice of a LARC and the decision to use prescription contraception.

These changes are large relative to experimentally estimated responses to financial costs. Using the randomized contraceptive subsidies in Bailey, Lang, Prettyman, et al. (2023) as a magnitude benchmark, the 4.1 percent increase in prescription contraceptive use generated by universal pharmacy access is comparable to the utilization response associated with roughly a \$29–\$41 subsidy, while the 11.6 percent increase under feasible LARC duration is comparable to roughly \$81–\$116.[10] The comparison indicates that the nonfinancial costs associated with obtaining and maintaining contraception are quantitatively meaningful relative to financial barriers to use.

[10]Using the subsidy amounts and estimated purchasing responses in Bailey, Lang, Prettyman, et al. (2023) yields a benchmark of approximately \$7–\$10 per percentage point of relative demand growth. I apply this relationship linearly to the counterfactual changes in prescription contraceptive use. These comparisons are illustrative, not structural price or welfare equivalents.

I next use the predicted changes in method choice to approximate their implications for pregnancy. I apply first-year typical-use failure rates to each contraceptive method (Centers for Disease Control and Prevention, 2013; Trussell, 2011) and assign an annual pregnancy probability of 12.09 percent to the outside option based on its estimated method composition in the NSFG. Universal pharmacy access predicts approximately 2,000 fewer pregnancies among Medicaid beneficiaries in an average year, while feasible LARC duration predicts approximately 26,100 fewer Medicaid pregnancies. Assuming that 47.9 percent of pregnancies result in a live birth (Bailey et al., 2022), these changes correspond to approximately 970 and 12,500 fewer births, respectively, in an average year.

I use these predictions to provide a back-of-the-envelope calculation of the corresponding implications for Medicaid spending (Table 10). Universal pharmacy access increases contraceptive spending by approximately $15.0 million annually through new short-acting prescriptions, net of reduced spending on methods from which women substitute. At an estimated Medicaid cost of $17,987 per birth in 2022 dollars (Sonfield et al., 2011), the predicted reduction in births corresponds to approximately $17.5 million in avoided expenditures and $2.5 million in net annual Medicaid savings. Feasible LARC duration increases contraceptive spending by approximately $192.7 million through additional LARC insertions, while the predicted reduction in births corresponds to approximately $224.4 million in avoided expenditures and approximately $31.8 million in net annual savings. Thus, both policies generate net Medicaid savings in the first year, with universal pharmacy access producing savings just above break-even and feasible LARC duration generating a larger reduction in expenditures.[11]

Together, the counterfactuals show that reducing the costs of obtaining and maintaining contraception can generate substantial changes in contraceptive use. These changes are meaningful relative to direct reductions in out-of-pocket prices and predict reductions in pregnancies among Medicaid beneficiaries, while the additional contraceptive spending may be offset by lower pregnancy-related expenditures. The access frictions documented in Section 2 also imply that these changes affect the resources required to obtain contraception. Pharmacist prescribing shifts routine prescribing toward a more geographically accessible setting and can reduce demand for clinic appointments, while longer LARC protection reduces the frequency with which continued contraceptive use requires clinical care. More broadly, the results distinguish making contraception formally available from making it easy to obtain and maintain, and show that these nonfinancial dimensions of access can be important determinants of demand even when monetary prices are small.

[11]These estimates are extrapolations from the model intended to illustrate fiscal magnitudes rather than a complete accounting of the budgetary or welfare consequences of either counterfactual. The calculations capture only pregnancies and Medicaid expenditures predicted in the first year following the change in contraceptive use. In particular, the feasible-duration calculation charges Medicaid the full upfront cost of additional LARC insertions without incorporating the pregnancy reductions associated with the additional years of protection those devices provide.

7 Competition and Contraceptive Duration

Though women value longer protection, firms may not immediately extend their products to the durations supported by clinical evidence. Manufacturers weigh the additional demand generated by longer protection against the reduction in replacement sales when each device lasts longer. Competition changes this tradeoff when a longer duration can attract consumers who would otherwise choose a rival's product.

Mirena entered the U.S. market in 2000 with five years of approved protection and faced no other hormonal IUD for more than a decade. The remaining long-acting market consisted of substantially differentiated products, including the copper IUD Paragard and, beginning in 2006, the contraceptive implant Implanon. Mirena's five-year label remained unchanged even as evidence accumulated that the device remained effective for longer. A randomized study had demonstrated efficacy through seven years as early as 1991 (Sivin et al., 1991), while a 2009 study examining extended use of Mirena documented continued levonorgestrel release and endometrial effects beyond five years, although its authors cautioned against recommending prolonged use on the basis of those findings alone (Hidalgo et al., 2009). By 2014, a review of the extended-use literature concluded that the available evidence supported seven years of Mirena use (Wu & Pickle, 2014).

Liletta entered in 2015 as another 52-mg levonorgestrel IUD and Mirena's only therapeutic substitute among LARCs. In the year subsequent to Liletta's entry, Bayer invested in the Mirena Extension Trial to establish efficacy beyond five years for FDA approval, with the investment in additional clinical evidence occurring only after the entry of a close rival.

What followed was a pattern of leapfrogging in approved duration. Liletta extended to four years in 2017, reached Mirena's five-year duration in 2018, and surpassed it with a six-year label in 2019. Mirena then extended for the first time in 2020, moving to six years, extended again to seven in 2021, before both products extended to eight years in 2022. Over five years, the two therapeutic substitutes moved from labels separated by two years to a common eight-year duration, with Liletta first closing and then reversing Mirena's initial duration advantage before Mirena caught up.

The experience of the other long-acting products provides an additional contrast. Paragard remained labeled for ten years despite clinical evidence supporting twelve years of use, while Nexplanon remained labeled for three despite evidence supporting five throughout the sample, with neither product facing a therapeutic substitute among LARCs. The persistence of these evidence-label gaps, despite extensions among Mirena and Liletta, is consistent with competition being particularly important between close therapeutic substitutes.

The timing alone does not establish that Liletta's entry caused Mirena's extensions, but it provides suggestive evidence consistent with a competitive response. Mirena remained at five years despite published evidence supporting longer use, Liletta entered as its first therapeutic substitute, Bayer subsequently invested in the trial necessary to support eight years of FDA-approved use, and the two products then leapfrogged one another in duration until both reached this ceiling. To examine the potential consequences of this competition for contraceptive use, I use the estimated demand system to simulate a market without Liletta under alternative assumptions about the duration Mirena would have offered in its absence.

7.1 Implications for Contraceptive Use

I use the estimated demand system to examine a counterfactual in which Liletta never enters the market. Liletta's entry introduced a therapeutic equivalent to Mirena, but it also preceded a series of extensions that increased Mirena's approved protection from five to eight years. The consequences of Liletta's absence therefore depend not only on how women would have substituted among existing contraceptives, but also on whether Mirena would have offered the same duration of protection without a close competitor. I consider three possible paths for Mirena to distinguish these possibilities. First, I remove Liletta while allowing Mirena to retain its observed extensions, isolating the contribution of Liletta's availability. Second, I consider the possibility that Mirena would not have extended without Liletta's entry, remaining at its original five-year approved duration. Finally, I allow Mirena to extend to six years, reflecting the possibility that Bayer would have sought to distinguish Mirena from its own five-year product, Kyleena, even without Liletta. Under this scenario, Mirena does not pursue its subsequent extensions to seven and eight years. All other products retain their observed characteristics and availability. Table 11 reports the counterfactual outcomes.

Removing Liletta alone has a relatively modest effect on aggregate contraceptive demand. When Mirena retains its observed extensions, annual LARC insertions decline by approximately 8,600 among Medicaid beneficiaries over 2020–2022, or 2.1 percent relative to the observed market. Much of Liletta's demand shifts to Mirena, whose annual insertions increase by approximately 18,600, while Kyleena and Skyla also gain users. The remaining decline reflects women who do not replace Liletta with another long-acting method.

The consequences are considerably larger when the absence of Liletta is accompanied by shorter approved protection from Mirena. When Mirena remains at five years, annual LARC insertions fall by approximately 58,500, a 14.3 percent decline relative to the observed market. This is nearly seven times the reduction associated with removing Liletta while preserving Mirena's observed duration. The difference reflects the loss of both an additional hormonal IUD and the longer protection that Mirena eventually offered. Although Kyleena and Skyla attract some of the women who would otherwise choose Mirena, neither provides more than five years of approved protection. The absence of a longer-duration hormonal IUD therefore reduces demand for long-acting contraception beyond the women who would have chosen Liletta itself.

Allowing Mirena to extend to six years substantially moderates this decline, but does not eliminate it. Under this scenario, annual LARC insertions fall by approximately 38,300, or 9.4 percent relative to the observed market. The initial extension preserves approximately 20,200 annual insertions compared with holding Mirena at five years. Nevertheless, annual insertions remain approximately 29,700 below the scenario in which Mirena retains its observed extensions. Even if Bayer had extended Mirena to distinguish it from Kyleena, the additional protection offered by its subsequent seven- and eight-year labels would still have contributed substantially to contraceptive demand.

Although some women substitute toward Kyleena and Skyla, the two remaining hormonal IUDs, shorter approved protection also reduces overall prescription contraceptive use. Across the three scenarios, approximately 7,600–52,000 fewer Medicaid women use prescription contraception annually, with the latter representing a 2.7 percent decline relative to the observed market. Some women shift toward short-acting methods, but others leave the

prescription contraceptive market altogether.

The decline in prescription contraceptive use and substitution to less effective methods also has consequences for reproductive outcomes. Applying the same contraceptive failure rates and pregnancy assumptions used in the earlier counterfactuals, the absence of Liletta generates approximately 1,000–6,700 additional unintended pregnancies and 470–3,200 additional births annually among Medicaid beneficiaries, depending on Mirena's approved duration (Table 11). Accounting for the additional births and changes in spending across individual contraceptive products, net Medicaid expenditures increase by approximately $7.1–$7.5 million annually across the three scenarios.

Had Liletta never entered the market, women might have faced a substantially different set of contraceptive options. Without Liletta itself or the longer durations subsequently offered by Mirena, fewer women would have chosen long-acting contraception, and some would have forgone prescription contraception altogether. The counterfactuals cannot establish which extensions Mirena would have pursued in Liletta's absence, but they reveal what is at stake in that decision. The direct loss of Liletta is relatively modest when Mirena retains its observed duration; the losses are considerably greater when the market also lacks the longer protection that Mirena eventually offered. The resulting increases in unintended pregnancies and births suggest that the consequences of improvements in contraceptive duration extend beyond the appeal of individual products.

8 Robustness

The main specification separates hormonal IUDs from other long-acting methods, allowing substitution to be stronger among products that provide the same method of contraception and differ primarily in characteristics such as duration and hormone dose. I alternatively place all long-acting methods in a single nest. The coefficient on duration falls from 0.314 to 0.106, while the nesting parameter increases from 0.743 to 0.904. This movement is expected when hormonal IUDs are combined with other LARCs, since more of the substitution generated by changes in duration is attributed to the common within-nest component. Under the preferred specification, an additional year of protection increases Mirena's share of the overall market by 0.80 percentage points and total LARC share by 0.54 percentage points. Under the broader LARC nest, Mirena's share increases by 1.14 percentage points, but total LARC share increases by only 0.20 percentage points, indicating that a larger portion of the response reflects substitution among long-acting methods. The estimated effect of pharmacist access also remains positive, with ϕ falling from 0.084 to 0.062. The estimated effect of pharmacist access is also positive when individual adopting jurisdictions or adoption cohorts are excluded, ranging from 0.038 to 0.124 across jurisdiction exclusions and from 0.074 to 0.153 across cohort exclusions, compared with 0.084 in the preferred specification. Appendix Table 16 reports these estimates.

I examine the sensitivity of the estimates to alternative instrument constructions in Appendix Table 15. The preferred instrument identifies whether a close rival introduced a product or extended its approved duration in the preceding year, while the alternative uses the maximum approved duration among competing hormonal IUDs in that year. Although both instruments exploit changes in rival product characteristics, they capture different di-

mensions of competition. The indicator identifies the timing of individual competitive events, which may coincide with changes in marketing, provider awareness, or other determinants of contraceptive demand. Lagged maximum rival duration instead reflects the level of protection already available from competing products, capturing both the duration offered by rivals and its proximity to the protection offered by a given product. It also allows earlier extensions to continue contributing variation beyond the year in which they occurred. This alternative may therefore be less sensitive to demand changes surrounding individual product introductions or extensions, although persistent changes in market conditions could still influence both rival characteristics and contraceptive choice. Comparing the two specifications allows me to examine whether the estimated effect of approved duration depends on the timing of competitive events or persists when the instrument reflects the relative protection offered by competing products.

Using lagged maximum rival duration produces a duration coefficient of $\beta = 0.384$ and a nesting parameter of $\rho = 0.821$, compared with 0.314 and 0.743 in the preferred specification. The estimated effect of duration remains positive under both constructions, with a somewhat larger coefficient when the instrument reflects the level of rival protection. I also examine whether individual duration extensions disproportionately influence the results by excluding each of the seven observed extensions in turn. The estimated duration coefficient remains positive in every specification, ranging from 0.277 to 0.471. Six exclusions produce estimates close to the preferred specification, while excluding Liletta's 2022 extension increases the duration coefficient to 0.471 and the nesting parameter to 0.921. Together, these exercises show that the positive estimated effect of approved duration persists across alternative representations of rival competition and is not driven by any single extension, although the magnitude is sensitive to Liletta's final extension.

Finally, I examine whether the omission of out-of-pocket payments affects the estimated valuation of approved duration. Medicaid coverage limits the prices women face, but differences in out-of-pocket costs may still influence contraceptive choice. Appendix Table 14 reports estimates from a specification that includes out-of-pocket payments as an exogenous control while retaining the preferred instruments for duration and within-nest share. The estimated duration coefficient changes only slightly, from 0.314 to 0.317, while the nesting parameter remains approximately 0.74. The coefficient on out-of-pocket payments is negative and statistically significant. The stability of the duration estimate suggests that the estimated effect of longer approved protection is not driven by the omission of observed contraceptive prices.

9 Conclusion

Contraceptive choice depends not only on the price of a method, but also on how frequently women must return to obtain, renew, or replace it. This paper studies two sources of variation in these costs: changes in the approved duration of long-acting contraceptives and state laws allowing pharmacists to prescribe short-acting methods. I find that women value reductions in both forms of hassle. An additional year of protection increases demand for hormonal IUDs, while pharmacist prescribing increases use of the methods it makes easier to obtain. These responses show that the time and effort required to maintain contraception

are economically meaningful determinants of demand even in a setting where consumers face little direct financial cost.

Longer approved protection increases the attractiveness of hormonal IUDs, but may also reduce the frequency with which women return for replacement devices. This creates a potential tension between the benefits of longer duration to consumers and manufacturers' incentives to extend it. Mirena retained a five-year label for roughly two decades, then began extending its approved duration following Liletta's entry and subsequent extensions. This pattern is suggestive of competition encouraging improvements in product characteristics that reduce the burden of maintaining contraception.

The counterfactuals show that these nonprice costs affect contraceptive participation, not merely substitution among products. In a market without Liletta where Mirena remains at its original five-year approved duration, annual Medicaid LARC insertions decline by approximately 58,500, and 52,000 fewer women use prescription contraception. Expanding pharmacist prescribing to all states similarly increases prescription contraceptive use, bringing approximately 70,800 additional Medicaid beneficiaries into the prescription contraceptive market each year. In both settings, changes in the convenience of contraception influence whether women use prescription methods at all.

These findings demonstrate that contraceptive access depends on more than financial coverage. The time and effort required to maintain protection can discourage contraceptive use even when out-of-pocket costs are low. Policies that reduce these burdens can increase participation, while the characteristics of durable contraceptives may themselves respond to competitive pressures. Understanding contraceptive access therefore requires attention not only to prices, but also to the policies and market conditions that determine how easily women can obtain and maintain effective protection.

Tables

Table 1: Contraceptive Methods: Access and Cost

	Frequency	Typical Use Rate	FDA-Approved Methods	How to get it	Max Annual Cost
Short-term methods					
Combination pill	Daily	7%	Generics available	Pharmacy	$600
Progestin-Only	Daily	7%	Generics available	Pharmacy	$600
Patch	Weekly	7%	Twirla, Xulane (generic)	Pharmacy	$600
Extended Use Pill	Daily	7%	Generics available	Pharmacy	$600
Ring	Monthly	7%	Annovera, NuvaRing, EluRyng (generic)	Pharmacy	$2,200
Injection	3-months	0.4%	Generics available	Doctor Visit	$600
LARCs					
Implant	3 years	0.05%	Nexplanon	Doctor Visit	$2,200
Hormonal IUD	3-8 years	0.1-0.4%	Mirena/Skyla/Liletta/Kyleena	Doctor Visit	$1,300
Copper IUD	10 years	0.8%	Paragard	Doctor Visit	$1,000

Notes: Source: FDA Birth Control Guide, Tschann & Soon (2015), Guttmacher Institute, and Pennington & Venator (2024). Typical use rates reflect the percentage of women experiencing unintended pregnancy within the first year of use. Maximum annual costs represent retail prices without insurance coverage. LARC = Long-Acting Reversible Contraceptive.

Table 2: Contraceptive Methods: Health Considerations

	RR VTE	**RR Depression**	**Amenorrhea (%)**	**Acne Treatment**
Short-term methods				
Combination pill	2.89	1.23	5	✓
Progestin-Only	1.0	1.34	10	
Patch	8.0	2.0	5	✓
Extended Use Pill	2.89	1.23	5	✓
Ring	6.5	1.6	3	✓
Injection	3.0	2.7	50	
LARCs				
Implant	2.0	2.1	22	
Hormonal IUD	1.0	1.5	6-20[a]	
Copper IUD	1.0	1.0	0	

[a] Varies by brand: Skyla/Kyleena (6%), Liletta/Mirena (20%)
RR = Relative Risk compared to non-users; VTE = Venous Thromboembolism
✓indicates method provides acne treatment benefits

Table 3: Product Entry and Approved versus Feasible Years of LARC Protection, 2015–2022

Product	Duration measure	Initial FDA approval year	Initial years	2015	2016	2017	2018	2019	2020	2021	2022
52-mg hormonal IUDs											
Mirena	Approved	2000	5	5	5	5	5	5	6	7	8
	Feasible			5	5	5	**6**	**7**	**8**	**8**	8
Liletta	Approved	2015	3	3	3	4	5	6	6	6	8
	Feasible			**4**	**5**	**6**	**7**	**8**	**8**	**8**	8
Lower-dose hormonal IUDs											
Kyleena	Approved	2016	5	–	5	5	5	5	5	5	5
	Feasible			–	5	5	5	5	5	5	5
Skyla	Approved	2013	3	3	3	3	3	3	3	3	3
	Feasible			3	3	3	3	3	3	3	3
Other long-acting methods											
Paragard	Approved	1984	10	10	10	10	10	10	10	10	10
	Feasible			**12**	**12**	**12**	**12**	**12**	**12**	**12**	**12**
Nexplanon	Approved	2011	3	3	3	3	3	3	3	3	3
	Feasible			**5**	**5**	**5**	**5**	**5**	**5**	**5**	**5**

Notes: Approved is the FDA-labeled period of contraceptive protection in force in the indicated calendar year. Feasible is the longest duration supported by the accumulated clinical evidence under the assumptions used in the counterfactual. **Bold** denotes a feasible duration strictly greater than the approved duration in that year. Initial FDA approval year and initial years refer to the named product, including approvals before 2015. A dash denotes a product not yet approved for marketing. Products are grouped by hormonal-IUD dose and other long-acting methods. Sources: FDA product approvals and extensions; Liletta ACCESS IUS trial (Creinin et al., 2022; Eisenberg et al., 2015); Mirena Extension Trial (Jensen et al., 2022); Paragard evidence (Ti et al., 2020; UNDP/UNFPA/WHO/World Bank Special Programme of Research, Development and Research Training in Human Reproduction, 1997); Nexplanon evidence (McNicholas et al., 2015, 2017).

Table 4: Descriptive Statistics across State-Year Contraceptive Markets, 2015–2022

	Mean	Median	Minimum	Maximum
Adjusted market size	285,442	174,239	8,435	2,771,394
Outside-option share	83.05	84.96	39.07	92.89
Prescription contraceptive share	16.95	15.04	7.11	60.93
Hormonal IUD share	1.92	1.69	0.01	7.53
Other LARC share	1.98	1.74	0.12	7.67
Short-acting share	13.06	11.58	4.52	52.75

Notes: Each observation is one state-year market; all statistics are unweighted across 384 markets in 48 jurisdictions over 2015–2022. Outside and prescription shares are percentages of the adjusted market of women, including the outside option. Nest shares are likewise percentages of that full market, not within-nest shares. Source: Medicaid State Drug Utilization Data; KFF Enrollment Reports, T-MSIS enrollment reports.

Table 5: Product-Level Utilization and Reimbursement in the Estimation Sample

Product	Full-market share (%)	Within-nest share (%)	Reimbursement ($)	N
Pill group 0	1.52	11.63	196	384
Pill group 1	0.93	7.40	567	384
Pill group 2	1.70	13.17	232	384
Pill group 3	0.80	5.67	164	382
Pill group 4	6.91	52.07	187	384
Patch	0.34	2.84	1,807	384
Ring	0.30	2.51	2,188	384
Injection	0.55	4.73	1,382	384
Mirena	1.44	74.70	898	384
Liletta	0.23	12.46	680	364
Kyleena	0.18	9.20	920	298
Skyla	0.12	6.50	698	375
Implant	1.60	78.45	871	384
Paragard	0.38	21.72	789	381

Notes: Shares are unweighted means over the state-year markets in which each product is observed. Full-market share divides product annualized choices/insertions by the adjusted market size, including the outside option. Within-nest share divides by choices within the product's own nest. Reimbursement is a claims-weighted pooled reimbursement measure (total reimbursement divided by total prescriptions). The pill-dose groups are the final model's five pill alternatives based on estrogen content. Source: Medicaid State Drug Utilization Data.

Table 6: Pharmacist Access and Use of Eligible Contraceptive Methods

Comparison	Eligible-method share (%)
State-years without pharmacist access	12.47
State-years with pharmacist access	12.60
Adopting jurisdictions, before adoption	11.68
Adopting jurisdictions, after adoption	11.88

Notes: Eligible methods include pills, patches, and rings. Shares are measured relative to the full contraceptive market, including the outside option. Access/no-access comparisons are unweighted state-year means across all 48 jurisdictions (296 without access and 88 with access); pre/post comparisons are state-level means among the 21 adopting jurisdictions. Differences are in percentage points. All comparisons are raw and descriptive, not adjusted estimates.

Table 7: First-Stage Estimates

	Approved years of protection	Log within-nest share
Excluded instruments		
Rival introduction or extension in prior year	0.4826	0.7651
	(0.0581)	(0.0935)
Close-dose competitor count	0.6403	-0.5529
	(0.0268)	(0.0806)
Included exogenous regressors		
Amenorrhea	1.3691	1.2315
	(0.0711)	(0.0501)
Pharmacist prescribing	-0.1986	-0.0859
	(0.0323)	(0.0273)
Method fixed effects	Yes	Yes
State fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Sanderson–Windmeijer conditional F	405	112
R^2	0.964	0.497
Observations	5,256	5,256

Notes: Each column reports one first-stage regression from the preferred nested-logit 2SLS specification, of the indicated endogenous variable (approved duration or log within-nest share) on both excluded instruments, the included exogenous regressors, and method, state, and year fixed effects. The first excluded instrument indicates whether a close hormonal-IUD rival (Mirena or Liletta) entered or extended its FDA-approved duration in the preceding year; the second counts other hormonal IUDs in the same dose tier available in the state-year. Pharmacist prescribing equals eligible method $\times$ active pharmacist-prescribing law. Standard errors in parentheses are clustered by state-year market (384 clusters). The reported statistic is the Sanderson–Windmeijer conditional F. Source: Medicaid State Drug Utilization Data; KFF Enrollment Reports; T-MSIS enrollment reports.

Table 8: Nested-Logit Demand Estimates

	Estimate	Standard error	95% confidence interval
Endogenous regressors			
Approved duration, β	0.3136	(0.1486)	[0.0224, 0.6048]
Within-nest log share, ρ	0.7430	(0.0988)	[0.5494, 0.9367]
Exogenous regressors			
Pharmacist prescribing, ϕ	0.0838	(0.0491)	[-0.0124, 0.1801]
Amenorrhea, γ	-0.1619	(0.1475)	[-0.4510, 0.1273]

Notes: Nested logit with method, state, and year fixed effects. Approved duration and log within-nest share are instrumented using a prior-year indicator for close-rival introduction or FDA-approved duration extension and a contemporaneous count of close-dose hormonal-IUD competitors. Pharmacist prescribing equals eligible method × active pharmacist-prescribing law. Standard errors are clustered by state-year market. N = 5,256 product-market observations across 384 state-year markets. Source: Medicaid State Drug Utilization Data; KFF Enrollment Reports; T-MSIS enrollment reports.

Table 9: Demand Counterfactuals: Annual Contraceptive Choices

	Baseline	Counterfactual	Change	Change (%)
Mirena +1 year: LARC insertions	388,415	442,266	53,851	13.9%
Universal pharmacy: eligible methods	1,271,491	1,356,133	84,642	6.66%
Feasible duration: LARC insertions	388,415	614,951	226,536	58.3%

Notes: Annual contraceptive choices are averaged over 2015–2022 and are flows, not stocks of users. Quantities use predicted shares multiplied by adjusted Medicaid market size. The Mirena exercise adds one year to Mirena's approved duration and re-solves all shares. Universal pharmacist prescribing activates the policy in all sample jurisdictions. The feasible-duration exercise uses the evidence-supported duration paths in Table 3. All counterfactuals use the preferred demand specification. Source: Medicaid State Drug Utilization Data; KFF Enrollment Reports; T-MSIS enrollment reports.

Table 10: Annual Medicaid Fiscal Counterfactuals

	Pregnancies averted	Births averted	Birth costs avoided ($M)	Device spending change ($M)	Short-acting spending change ($M)	Net savings ($M)
Universal pharmacy access	2,036	973	17.50	-2.34	17.38	2.46
Feasible duration	26,103	12,477	224.43	198.60	-5.94	31.77

Notes: Pregnancy probabilities are 0.1209 for the outside option, 0.089 for short-acting methods, 0.002 for hormonal IUDs, 0.008 for copper IUDs, and 0.0005 for implants. Births per pregnancy are 0.478; Medicaid cost per birth is $17,987. Positive spending changes represent additional expenditure, and negative changes represent savings. Net savings equal avoided birth costs minus the changes in device and short-acting spending. The accounting charges additional devices in the insertion year and credits one year of avoided births. Sources: Medicaid State Drug Utilization Data; KFF Enrollment Reports; T-MSIS enrollment reports; Bailey, Lang, Vortuba, and Bartík, 2023; Finer and Zolna, 2016; Sonfield et al., 2011; Trussell, 2011.

Table 11: Contraceptive Use, Reproductive Outcomes, and Medicaid Expenditures without Liletta

	Observed	No Liletta: Mirena duration		
	market	Observed	Six years	Five years
Contraceptive use (average annual levels)				
Total LARC insertions	408,842	400,268	370,526	350,348
Mirena	134,319	152,953	91,672	44,773
Kyleena	23,625	26,929	50,182	68,994
Skyla	9,198	10,503	18,289	25,810
Liletta	31,970	0	0	0
Any prescription contraceptive users	1,900,850	1,893,228	1,866,693	1,848,857
Annual change relative to observed market				
LARC insertions	—	−8,574	−38,316	−58,494
Prescription contraceptive users	—	−7,621	−34,157	−51,993
Unintended pregnancies	—	+989	+4,423	+6,748
Births	—	+473	+2,114	+3,225
Annual change in Medicaid expenditures (millions of dollars)				
Birth-related expenditures	—	+8.5	+38.0	+58.0
Contraceptive expenditures	—	−1.1	−30.5	−50.9
Net Medicaid expenditures	—	**+7.5**	**+7.5**	**+7.1**

Notes: All values are Medicaid annual flows averaged over 2020–2022. Changes are relative to the observed market. The no-Liletta scenarios respectively retain Mirena's observed duration extensions, cap its duration at six years, or hold it at five years. Reproductive outcomes use the method-specific contraceptive failure rates and pregnancy assumptions in Table 10, 0.478 births per unintended pregnancy, and $17,987 in Medicaid expenditures per birth. Contraceptive spending uses product-specific device costs (Mirena $944, Liletta $720, Kyleena $925, Skyla $732, copper IUD $820, implant $930) and $236.70 annually for short-acting methods. Net Medicaid expenditures incorporate both additional birth-related costs and changes in contraceptive spending.

Figures

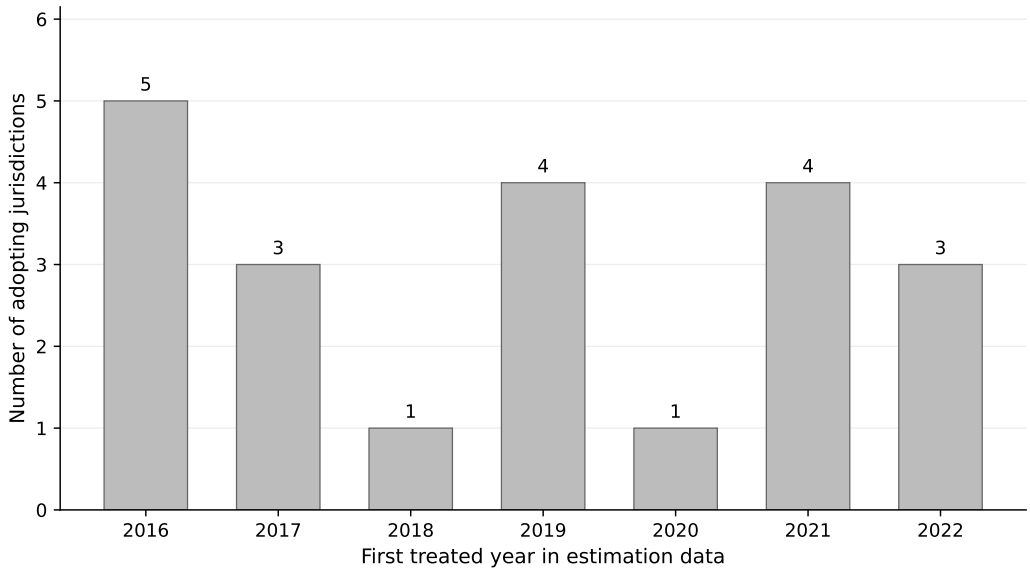


Figure 1: Adoption of pharmacist prescribing, 2016–2022

Notes: Bars show the number of jurisdictions first coded as allowing pharmacists to prescribe eligible contraception in each year. The sample contains 48 state-level jurisdictions, including the District of Columbia. Twenty states and DC adopt during the sample.

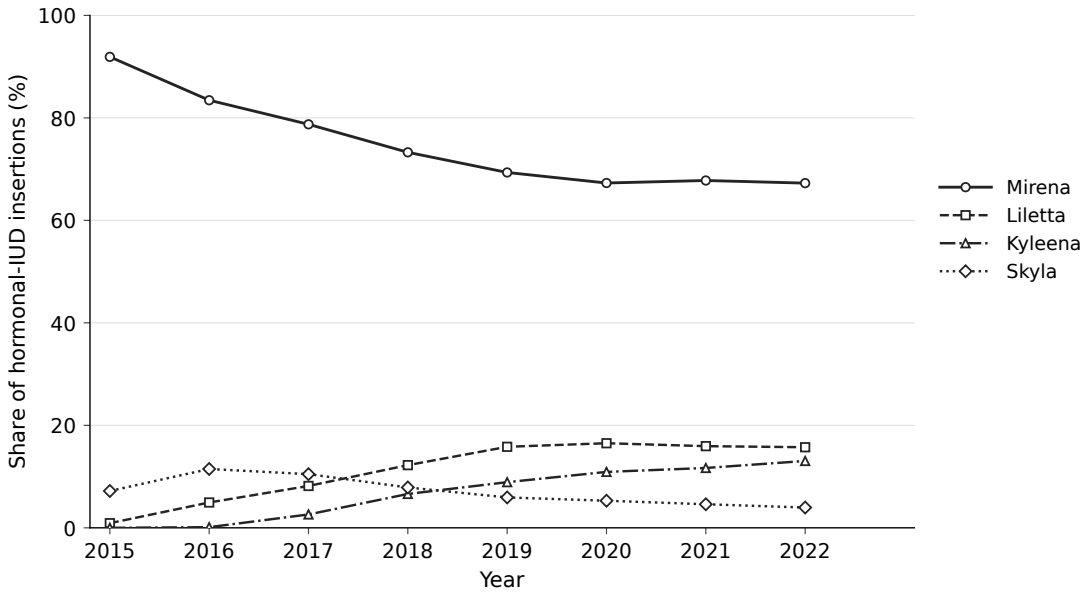


Figure 2: Product shares of Medicaid hormonal-IUD insertions, 2015–2022

Notes: Each series shows the percentage of Medicaid hormonal-IUD insertions attributable to the indicated product in a given year. The denominator is total insertions across the four hormonal-IUD products, not all contraceptive users or all Medicaid women. Line patterns and point markers distinguish products in grayscale. *Source:Medicaid State Drug Utilization Data; KFF Enrollment Reports, T-MSIS enrollment reports.* .

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A Data Construction

This appendix states the steps that build the product-level contraceptive market from the raw administrative records.

Source and panel. The raw data are the Medicaid State Drug Utilization Data, which report prescriptions, units reimbursed, and dollars reimbursed by the state and by other payers for each drug code in each state and calendar quarter, separately for fee-for-service and managed care. Records are summed across drug codes mapping to the same product and across the two delivery systems, then across quarters, to give a product–state–year panel for calendar years 2015 through 2022. Three jurisdictions with discontinuous contraceptive reporting are dropped, leaving forty-eight jurisdictions over eight years (384 state-year markets) and, after product aggregation, 5,256 product-market observations across fourteen alternatives.

Products. Drug codes are mapped to the fourteen alternatives by active ingredient, method, and, for oral contraceptives, estrogen dose. The four hormonal intrauterine devices (Mirena, Liletta, Kyleena, Skyla) are kept separate; the copper device, the implant, the injection, the patch, and the ring are each a single alternative; oral contraceptives are grouped into five estrogen-dose categories. A product is included in a state-year only in years with positive Medicaid utilization there, so that the post-launch appearance of Liletta (2015) and Kyleena (2016) is treated as non-availability rather than as a zero share.

Quantity standardization. Raw volume is converted to the number of women supplied for one year. Oral-pill units are divided by 365; patch and monthly-ring prescriptions are divided by 12; injection units are divided by the 150-mg standard dose; each intrauterine device and implant counts as one insertion. These counts are the share numerators and are annual flows.

Market size. Let E_{st} be the number of female Medicaid enrollees ages 18–45 in state s and year t, obtained by multiplying compiled total enrollment by the reproductive-age female share from the Current Population Survey. Women still protected by a prior long-acting insertion are removed from the choosing population, following Armitage (2022). With $q_{js\tau}$ insertions of product j in state s and year τ, and approved protection length $D_{j\tau}$ at the time of insertion, the stock still covered in year t is

$$S_{st} = \sum_{j \in \mathcal{L}} \sum_{\tau \,:\, \tau < t \leq \tau + D_{j\tau}} q_{js\tau}, \tag{1}$$

where $\mathcal{L}$ includes hormonal IUDs, the copper IUD, and the implant. The continuation rate is set to one, so each cohort is assumed to remain protected for the full approved duration. The adjusted market size is

$$M_{st} = E_{st} - S_{st}. \tag{2}$$

Insertion cohorts that adopted before 2015 are imputed by applying observed insertion rates to earlier enrollment.

Table 12 illustrates the market-size adjustment aggregated across all 384 state-year markets. Continuing LARC users account for approximately 13 million women across these markets, reducing the aggregate market size by 11.83 percent.

Table 12: Market-Size Adjustment for Continuing LARC Users

	Continuing users removed	Remaining market size
Initial market size	—	109,609,722
Continuing hormonal IUD users	6,729,433	102,880,289
Continuing copper IUD users	2,839,174	100,041,115
Continuing implant users	3,402,943	96,638,172
Total continuing LARC users	12,971,550	
Adjusted market size		96,638,172

Notes: The table reports the reconstructed market-size adjustment aggregated across 384 state-year markets from 2015 through 2022. Continuing users are identified from historical LARC insertions and approved protection durations at insertion, assuming a continuation rate of one. The aggregate reduction is 11.83 percent, calculated as the total continuing-LARC stock divided by the initial market size.

Shares and the dependent variable. Each product's share is its standardized count divided by M_t; the outside share is the market residual not accounted for by the fourteen alternatives. Products are assigned to three nests—the hormonal intrauterine devices, the other long-acting methods (copper device and implant), and the short-acting methods (pills, patch, ring, injection). Observed shares are inverted to the mean-utility index used as the dependent variable in the demand model,

Characteristics. Approved duration is the FDA-labeled length of protection in force in each product-year (Table 3); short-acting methods take the fraction of a year covered by one administration, namely 1/365 for pills, 1/52 for the patch, 1/12 for the ring, and 1/4 for the injection. A second duration measure, the length the accumulated clinical evidence already supported, is dated from each product's trial and post-approval milestones and is used only in the feasible-duration counterfactual. The amenorrhea characteristic is a binary indicator for the methods that commonly suppress menstruation—the two higher-dose hormonal devices, the implant, and the injection. Pharmacist access is one for pharmacy-dispensed methods in states and years whose scope-of-practice law permits pharmacist prescribing, dated from statutory effective dates and absorbing thereafter; twenty states and the District of Columbia adopt between 2016 and 2022.

B Additional Demand Results

B.1 Sensitivity of the pharmacist-access coefficient

Table 13 re-estimates the preferred specification while excluding adopting jurisdictions and adoption cohorts in turn. Dropping each of the twenty-one adopting jurisdictions, the coefficient ranges from 0.038 to 0.124 around its preferred value of 0.084; dropping each adoption cohort, it ranges from 0.074 to 0.153. The estimated coefficient remains positive across these exclusions.

Table 13: Sensitivity of the pharmacist-access coefficient

Specification or exclusion	Coefficient	Std. error
Preferred specification	0.0838	0.0491
Leave-one-jurisdiction-out (21 adopters)		
Range across exclusions	[0.0383, 0.1241]	
Minimum (exclude jurisdiction ID 8)	0.0383	0.0437
Maximum (exclude jurisdiction ID 6)	0.1241	0.0502
Leave-one-cohort-out		
Exclude 2016 cohort (5 jurisdictions)	0.1534	0.0648
Exclude 2017 cohort (3)	0.0854	0.0580
Exclude 2018 cohort (1)	0.0383	0.0437
Exclude 2019 cohort (4)	0.0848	0.0524
Exclude 2020 cohort (1)	0.0906	0.0498
Exclude 2021 cohort (4)	0.0886	0.0536
Exclude 2022 cohort (3)	0.0735	0.0503

Notes: Each exclusion re-estimates the preferred instrumental-variables demand model, including amenorrhea, pharmacist access, and method, state, and year fixed effects. Standard errors are clustered by state-year market. Jurisdiction identifiers have not been matched to names; the sample changes with each exclusion. The preferred sample contains 5,256 product-market observations.

B.2 Including out-of-pocket payments

I reestimate the demand model including out-of-pocket payments to examine whether omitting price affects the estimated valuation of approved duration and pharmacist access. I treat price as exogenous in this specification. Medicaid contraceptive coverage limits patient cost sharing, and the remaining payments are largely governed by federal coverage requirements and state Medicaid policies rather than prices chosen by manufacturers in response to demand. This institutional setting reduces the concern that unobserved demand conditions influence the prices women face, which would otherwise motivate instrumenting for price.

Table 14 reports the results. Including price leaves the estimated coefficient on approved duration nearly unchanged, increasing from 0.314 to 0.317. The pharmacist-access coefficient remains 0.084, and the nesting parameter changes from 0.743 to 0.741. The coefficient on out-of-pocket payment is negative and statistically significant.

Table 14: Demand estimates including out-of-pocket payments

	Preferred specification	Including price
Approved duration	0.31358	0.31697
	(0.14857)	(0.14871)
Nesting parameter	0.74303	0.74133
	(0.09881)	(0.09894)
Pharmacist access	0.08384	0.08407
	(0.04911)	(0.04887)
Amenorrhea	-0.16188	-0.16415
	(0.14751)	(0.14772)
Out-of-pocket payment	—	-0.00032
		(0.00011)
Method fixed effects	Yes	Yes
State fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Sanderson–Windmeijer F, duration	405	401
Sanderson–Windmeijer F, within-nest share	112	112
Observations	5,256	5,256
State-year clusters	384	384

Notes: The second column adds out-of-pocket payment as an exogenous control to the preferred nested-logit demand model. Both specifications treat approved duration and log within-nest share as endogenous, instrumented with the preceding-year close-rival introduction-or-extension indicator and the contemporaneous close-dose hormonal-IUD competitor count. Both include method, state, and year fixed effects, amenorrhea, and pharmacist prescribing for eligible methods. Standard errors clustered by state-year market are in parentheses. The price variable is the product-market share-weighted, nonnegative difference between price and Medicaid-covered service cost, measured in dollars. Sanderson–Windmeijer conditional F statistics are reported for both endogenous regressors.

B.3 Alternative instruments and nesting

The preferred duration instrument indicates whether a close rival entered the market or extended its approved duration in the preceding year, while the second instrument counts other same-dose-tier hormonal-IUD competitors in the current state-year. Table 15 examines alternative constructions of the excluded instruments. Replacing the lagged rival-event indicator with the lagged maximum approved duration among competing hormonal IUDs produces a positive duration coefficient of 0.384, compared with 0.314 in the preferred specification. Both instruments exploit rival product characteristics, so the comparison does not eliminate the underlying exclusion-restriction concern. An additional specification using the rival-event and lagged-maximum-duration instruments together has considerably weaker conditional first stages.

Panel B excludes each of the seven observed duration extensions in turn. The estimated duration coefficient remains positive throughout. Six exclusions produce estimates between 0.277 and 0.330; excluding Liletta's 2022 extension increases the duration coefficient to 0.471 and the nesting parameter to 0.921. Table 16 also places all long-acting methods in a single

nest. This reduces the duration coefficient from 0.314 to 0.106 and increases the nesting parameter from 0.743 to 0.904.

Table 15: Instrument robustness and leave-one-duration-extension-out estimates

Specification	Duration β	Nesting ρ	Pharmacist ϕ	Amenorrhea γ	SW F: duration	SW F: nest	N
Alternative excluded instruments							
Rival event + close-dose count (preferred)	0.3136 (0.149)	0.7430 (0.099)	0.0838 (0.049)	-0.1619 (0.148)	405	112	5,256
Lagged rival maximum duration + close-dose count	0.3835 (0.171)	0.8211 (0.078)	0.1065 (0.053)	-0.3900 (0.202)	634	397	5,256
Rival event + lagged rival maximum duration	0.0077 (0.195)	0.9330 (0.097)	0.0312 (0.054)	0.0561 (0.189)	23	47	5,256
Exclude one FDA duration extension							
Mirena 2020 (5 to 6 years)	0.2929 (0.144)	0.7186 (0.109)	0.0778 (0.049)	-0.1038 (0.155)	311	—	5,208
Mirena 2021 (6 to 7 years)	0.3014 (0.139)	0.7288 (0.105)	0.0764 (0.047)	-0.1117 (0.108)	500	—	5,208
Mirena 2022 (7 to 8 years)	0.2768 (0.130)	0.7017 (0.113)	0.0640 (0.046)	-0.0131 (0.088)	340	—	5,208
Liletta 2017 (3 to 4 years)	0.3304 (0.158)	0.7640 (0.091)	0.0867 (0.050)	-0.2371 (0.192)	357	—	5,212
Liletta 2018 (4 to 5 years)	0.3096 (0.148)	0.7371 (0.102)	0.0822 (0.049)	-0.1431 (0.159)	302	—	5,209
Liletta 2019 (5 to 6 years)	0.2904 (0.138)	0.7141 (0.110)	0.0774 (0.048)	-0.0626 (0.121)	422	—	5,210
Liletta 2022 (6 to 8 years)	0.4712 (0.224)	0.9212 (0.049)	0.1140 (0.055)	-0.5966 (0.330)	235	—	5,208

Notes: All models use method, state, and year fixed effects, amenorrhea, and pharmacist prescribing. Standard errors in parentheses are clustered by state-year market. The preferred event instrument indicates whether a close rival entered or extended approved duration in the previous year; the rival maximum-duration instrument uses the previous year's maximum approved duration among other hormonal IUDs. The close-dose count measures other same-dose-tier competitors in the current market. Sanderson–Windmeijer conditional F statistics are reported where supplied; the within-nest first-stage statistics for extension exclusions were not provided. The rival-event plus lagged-maximum-duration specification has substantially weaker conditional first stages. Source: preferred-specification robustness reruns.

Table 16: Demand robustness: alternative nesting structure

Specification	Duration β	Nesting ρ	Pharmacist ϕ	Amenorrhea γ	SW F (dur./nest)	Mirena/LARC effect (pp)
Preferred hormonal-IUD nest	0.3136 (0.149)	0.7430 (0.099)	0.0838 (0.049)	-0.1619 (0.148)	405 / 112	0.803 / 0.544
All long-acting methods in one nest	0.1060 (0.065)	0.9041 (0.039)	0.0622 (0.044)	-0.0449 (0.074)	558 / 48	1.142 / 0.201

Notes: Clustered standard errors in parentheses. The last column reports full-market percentage-point changes in Mirena and total LARC shares following a one-year extension of Mirena's approved duration. Both models include amenorrhea, pharmacist access, and method, state, and year fixed effects. Preferred $N = 5{,}256$ across 384 state-year markets.